

INFINITY FITNESS STUDIOS

Analysis & Segmentation Methodology

Member Retention, Utilization & Operational Analytics
December 2025 - February 2026

Document Control	Details
Document ID	IFS-ASM-001
Version	1.0
Role	Data Analyst
Purpose	Document the analytical workflow used to prepare, segment and interpret membership, attendance, renewal, utilization and operational data.
Confirmed Tools	Python, SQL and Power BI
Source	Approximately one year of Excel-based business data
Method Evidence	Cleaning, EDA, cohort analysis, membership-duration segmentation, join/exit comparison, attendance-behavior analysis, calculations, visualization and Power BI filtering were confirmed for the engagement.
Control	Exact client values, source-field names, SQL statements, historical thresholds and formulas are not reconstructed where they are not retained.

1. Analytical Objective

The analysis was designed to move the studio from fragmented operational data toward a consolidated understanding of member behavior and business utilization. The main analytical questions concerned retention and attrition, attendance behavior, membership renewals, plan performance, peak and underutilized periods, trainer utilization and commercial performance.

2. End-to-End Methodology

Stage	Method	Purpose
1. Requirements Discovery	Calls, WhatsApp discussions, meetings, review of existing reports and Excel sheets with the gym manager.	Understand management questions and available evidence.
2. Data Preparation	Clean and structure the Excel-based analytical data before analysis.	Reduce data-quality problems that could distort member, attendance, payment or utilization results.
3. Exploratory Data Analysis	Use Python and analytical querying to inspect patterns, distributions and relationships.	Understand the data before defining business views.
4. Member Segmentation	Segment members by joining period, membership duration, plan and relevant activity behavior.	Compare member groups rather than relying only on aggregate totals.
5. Cohort Analysis	Group members by joining month and track subsequent attendance/retention behavior.	Identify when engagement changes after acquisition.
6. Attendance & Utilization Analysis	Analyze attendance by hour/day and membership usage.	Identify high-use and underutilized periods.
7. Renewal / Plan Analysis	Compare renewal behavior and plan performance where supported by data.	Support pricing, promotion and re-engagement discussions.
8. Trainer Analysis	Review sessions, bookings, workload, member assignments and attendance.	Provide trainer-utilization visibility.
9. Visualization	Build the Power BI dashboard from scratch and use filtering to investigate segments and trends.	Make analytical findings accessible to management.
10. Recommendation Development	Translate evidence into pricing and operational recommendations.	Move from descriptive reporting to decision support.

3. Data Preparation Principles

- Review the supplied Excel data before calculating business metrics.
- Standardize date fields needed for joining, renewal, attendance, payment and period analysis.
- Resolve or flag duplicates where they could inflate member, attendance, payment, booking or session counts.
- Standardize membership-plan labels before comparing plan behavior.
- Check key identifiers before linking member, membership, attendance, payment, trainer and session data.
- Treat missing renewal, cancellation or attendance evidence carefully; missing information is not automatically churn.
- Validate numeric fields before revenue/expense aggregation.

4. Exploratory Data Analysis (EDA)

EDA was used to understand member activity and business patterns before dashboarding. Confirmed work included cleaning, calculations, segmentation and visualization in Python, with SQL also used in the analytical workflow. Because the exact historical SQL operations are not retained, this document does not invent query structures or database objects.

EDA Area	Analytical Question
Membership	How are members distributed across available plan types and joining periods?

Attendance	How does attendance vary over time and across members?
Renewals	What renewal behavior is visible across relevant member/plan groups?
Utilization	Which hours/days show comparatively high or low activity?
Trainer Activity	How do sessions, bookings, workload, assignments and attendance vary?
Commercial	What revenue and expense patterns are visible in the supplied data?

5. Segmentation Framework

Segment	Basis	Why It Matters
Joining Cohort	Joining month	Separates members by acquisition period and supports longitudinal retention analysis.
Membership Duration	Elapsed membership duration / tenure	Helps identify whether engagement changes as membership progresses.
Plan Type	Monthly, 3-month, 6-month, annual, personal training	Supports comparison of plan behavior and renewal patterns.
Activity / Attendance	Observed attendance behavior	Helps distinguish stronger engagement from declining or stopped attendance.
Renewal Behavior	Renewed versus not renewed where evidence supports classification	Supports renewal and re-engagement analysis.
Time of Use	Hour/day or time band	Supports peak and underutilized-period analysis.
Trainer Activity	Sessions, bookings, workload, assignments and attendance	Supports trainer-utilization analysis.

6. Cohort Analysis Method

Members were grouped by joining month and their subsequent attendance/retention behavior was tracked across later periods. This prevents newer and older members from being mixed into a single retention number and makes engagement decay easier to observe.

Step	Cohort Method
1	Derive the member's joining month from the joining date.
2	Assign each eligible member to the corresponding joining-month cohort.
3	Track subsequent attendance and/or retained-membership evidence across later periods.
4	Compare the number or share of engaged/retained members with the original eligible cohort.
5	Inspect the pattern across cohorts and periods for consistent engagement decline or retention differences.
6	Use Power BI filters and supporting Python analysis to investigate relevant segments.

7. Key Pattern Identified

The analysis identified declining member engagement following the New Year period, with the decline becoming visible approximately one month later. The finding was investigated through joining-month cohorts, membership-duration segmentation, attendance behavior and member activity over time.

The public case study deliberately avoids turning this into an unsupported universal rule such as 'all members churn after three months.' Exact cohort counts, percentages and confidential client values are not published.

8. Attrition & Retention Logic

Concept	Methodological Treatment
Retention	Members continuing their membership over the relevant analysis period.

Attrition / Churn	Member disengagement evidenced through stopped attendance and/or membership cancellation under the project definition.
Stopped Attendance	Used as an engagement signal; exact inactivity threshold is not retained and therefore is not fabricated.
Cancellation	Used as attrition evidence where cancellation information is available.
Renewal	Analyzed separately using membership renewal evidence.

9. Utilization Methodology

Analysis	Method	Interpretation
Attendance by Time	Aggregate valid check-ins by hour/day or appropriate time band.	Identify relative demand across operating periods.
Peak Periods	Compare attendance concentration across time periods.	Evening activity emerged as an important utilization period.
Underutilized Periods	Identify comparatively lower-use periods from the same attendance pattern.	Provide management visibility into periods with lower activity.
Membership Usage	Review member attendance frequency/activity over selected periods.	Support engagement and utilization interpretation.

10. Plan & Renewal Analysis

Membership behavior was considered across monthly, 3-month, 6-month, annual and personal-training offerings where corresponding records were available. Renewal and engagement evidence informed recommendation areas including renewal discounts, longer-duration plan incentives, inactive-member re-engagement offers and quarterly promotions.

The methodology does not claim that a particular plan produced a specific uplift or financial result because exact historical values are not retained for public disclosure.

11. Trainer Utilization Methodology

Input	Analytical Use
Sessions	Compare recorded trainer activity.
Class Bookings	Understand demand associated with trainer/class activity.
Workload	Assess relative activity using the available workload indicators.
Member Assignments	Understand trainer-member allocation where recorded.
Attendance	Compare realized member activity with trainer/session information.

No single composite trainer-utilization formula or weighting is asserted because the exact historical formula is not retained.

12. Recommendation Logic

Finding Area	Decision Translation
Renewal behavior	Consider targeted renewal discounts.
Plan duration	Consider incentives that encourage longer-duration memberships.
Inactive members	Consider re-engagement offers.
Promotional cadence	Consider quarterly offers.
Peak / underutilized periods	Use utilization evidence to support operating decisions.
Trainer activity	Use workload, booking and attendance visibility to support allocation discussions.

13. Qualitative External Context

Publicly visible information from the studio's page and comments was reviewed as qualitative context for operational observations. This material was not treated as equivalent to structured internal transaction or attendance data. Any public-comment evidence should be separated from quantitative findings when presenting the case study.

14. Analytical Validation

- Reconcile dashboard totals and filtered views to the prepared analytical data.
- Check cohort membership so a member is assigned to the correct joining period.
- Check date and time transformations before trend and peak-hour analysis.
- Review plan standardization before plan-level comparisons.
- Validate that joins do not duplicate attendance, payments, sessions or members.
- Distinguish qualitative observations from calculated quantitative findings.
- Do not convert missing historical detail into fabricated thresholds, formulas or impact values.

15. Tool Responsibilities

Tool	Confirmed Role / Safe Documentation
Excel	Source format and analytical working material supplied/reviewed during the engagement.
Python	Cleaning, EDA, cohort analysis, segmentation, calculations and visualization.
SQL	Confirmed as part of the analytical toolset; exact historical querying/aggregation activity remains to be verified before detailed claims are added.
Power BI	Dashboard built from scratch; used for analytical visualization and filtering.

16. Outputs of the Methodology

- Retention and attrition analysis
- Joining-month cohort analysis
- Membership-duration and activity segmentation
- Attendance, peak-period and underutilized-period analysis
- Renewal and membership-plan analysis
- Trainer-utilization analysis
- Commercial analysis using available revenue/expense information
- Power BI dashboard
- Excel analysis, Python notebook, presentation, report, written recommendations and live walkthrough

17. Known Limitations

Limitation	Treatment
Exact source row/member counts not retained	Do not publish invented counts.
Exact inactivity threshold not retained	Describe stopped attendance as an attrition signal without fabricating a threshold.
Exact SQL implementation not retained	Keep SQL at tool level until specific activity is recalled/verified.
Exact dashboard page structure not retained	Do not reconstruct page names/count as historical fact.

Exact financial/retention impact not retained or confidential	Use qualitative business value only.
Client data restricted	Use sanitized methodology and portfolio-safe visuals rather than raw data.